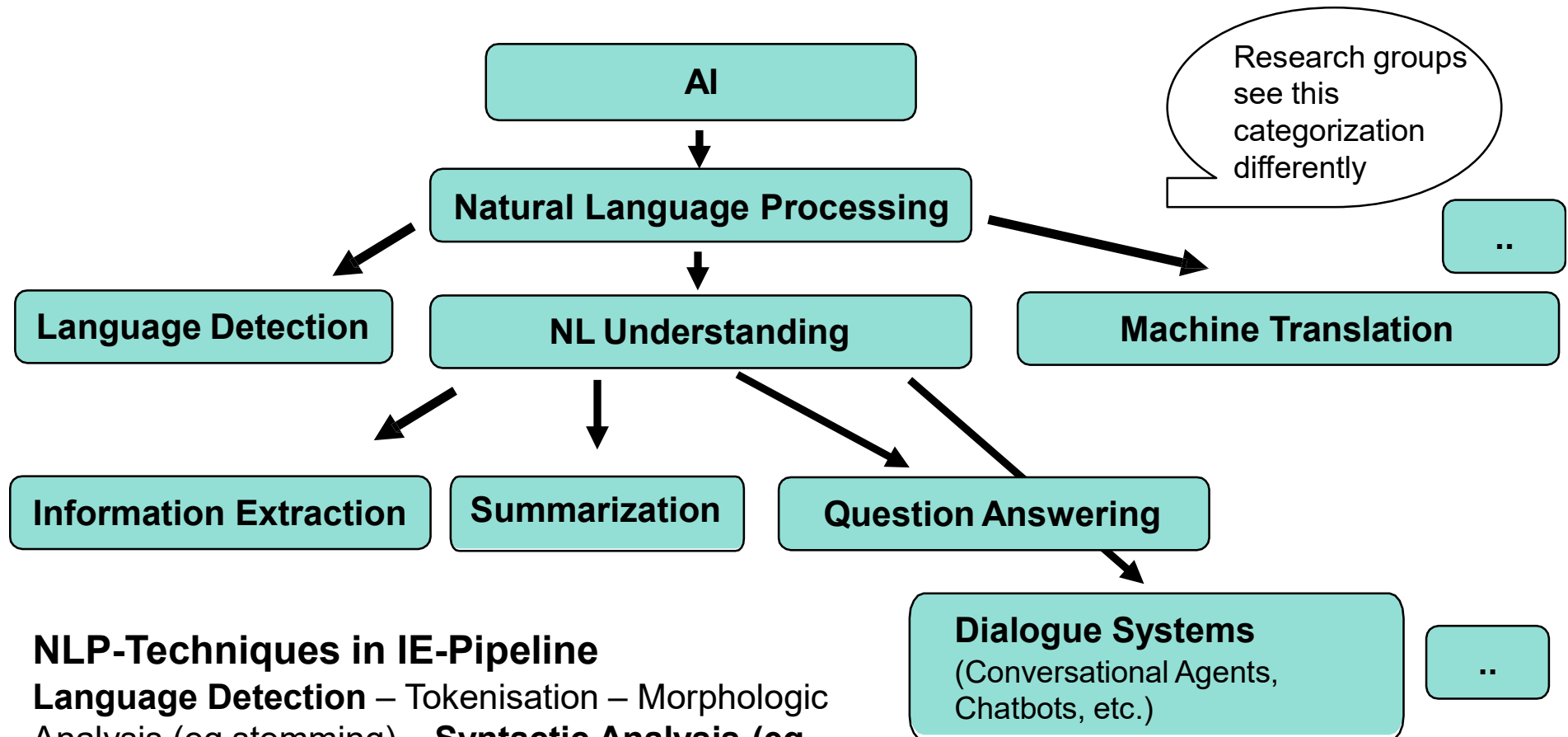


Contents – Lecture 5

- 1) Information Retrieval (IR) & Information Extraction (IE) in a Nutshell
- 2) Web Search – From Crawlers to Ranking
- 3) Search Engine Optimization (SEO), Site Search
- 4) Web Scraping / Extraction
- 5) Social Media Mining**
 - ◆ Natural Language Processing (NLP)
 - ◆ Opinion Mining / Sentiment Analysis
- 6) Question Answering and Dialogue Systems
- 7) Search User Interfaces (?)

Natural Language Processing (NLP)

[D. Jurafsky, J. Martin: Speech and Language Processing; 2009] current version as ONLINE BOOK!!



NLP-Techniques in IE-Pipeline

Language Detection – Tokenisation – Morphologic Analysis (eg stemming) – **Syntactic Analysis (eg POS-Tagging)** – Semantic Analysis (eg based on knowledge base)

Natural Language Processing

Part-of-Speech-Tagging (POS-Tagging) (1/3)

■ History: Dionysius Thrax of Alexandria (~ 100B.C.) [?]

- 8 parts-of-speech: noun, verb, pronoun, preposition, adverb, conjunction, participle, and article
- Basis for POS for the next 2000 years
- Current lists of parts-of-speech have more word classes, e.g. C7 tagset: 146

■ Definition POS-Tagging

- Classification of words of a corpus according to POS-tagset; annotation with POS-tagset entries; word category “disambiguation”

■ Why POS-Tagging?

- Advice for meaning: e.g.: content => noun: CONtent, adjective conTENT (in this example also pronunciation)
- Advice what words are likely to occur in its vicinity, e.g.: personal pronouns are likely to be followed by a verb

Natural Language Processing

Part-of-Speech-Tagging (POS-Tagging) (2/3)

■ Approaches

- Rule-based-tagging (hand-written-rules)
- Statistical methods: hidden markov model tagging, maximum entropy tagging
- Others

■ Some POS-Taggers

- Stanford POS-Tagger
- Brill tagger
- STTS (Stuttgart-Tübingen Tagset): für deutsche Sprache
- POS-Tagger for Twitter: e.g.: GATE Twitter PoS tagger

Natural Language Processing

Part-of-Speech-Tagging (POS-Tagging) (3/3)

nlp.stanford.edu/~jurafsky/slp3/9.pdf

THE PENN TREEBANK PART-OF-SPEECH TAGSET

Tag	Description	Example	Tag	Description	Example
CC	coordin. conjunction	<i>and, but, or</i>	SYM	symbol	<i>+, %, &</i>
CD	cardinal number	<i>one, two</i>	TO	"to"	<i>to</i>
DT	determiner	<i>a, the</i>	UH	interjection	<i>ah, oops</i>
EX	existential 'there'	<i>there</i>	VB	verb base form	<i>eat</i>
FW	foreign word	<i>mea culpa</i>	VBD	verb past tense	<i>ate</i>
IN	preposition/sub-conj	<i>of, in, by</i>	VBG	verb gerund	<i>eating</i>
JJ	adjective	<i>yellow</i>	VBN	verb past participle	<i>eaten</i>
JJR	adj., comparative	<i>bigger</i>	VBP	verb non-3sg pres	<i>eat</i>
JJS	adj., superlative	<i>wildest</i>	VBZ	verb 3sg pres	<i>eats</i>
LS	list item marker	<i>1, 2, One</i>	WDT	wh-determiner	<i>which, that</i>
MD	modal	<i>can, should</i>	WP	wh-pronoun	<i>what, who</i>
NN	noun, sing. or mass	<i>llama</i>	WP\$	possessive wh-	<i>whose</i>
NNS	noun, plural	<i>llamas</i>	WRB	wh-adverb	<i>how, where</i>
NNP	proper noun, sing.	<i>IBM</i>	\$	dollar sign	<i>\$</i>
NNPS	proper noun, plural	<i>Carolinas</i>	#	pound sign	<i>#</i>
PDT	predeterminer	<i>all, both</i>	"	left quote	<i>' or "</i>
POS	possessive ending	<i>'s</i>	"	right quote	<i>' or "</i>
PRP	personal pronoun	<i>I, you, he</i>	(	left parenthesis	<i>[, (, {, <</i>
PRP\$	possessive pronoun	<i>your, one's</i>	)	right parenthesis	<i>],), }, ></i>
RB	adverb	<i>quickly, never</i>	,	comma	<i>,</i>
RBR	adverb, comparative	<i>faster</i>	.	sentence-final punc	<i>. ! ?</i>
RBS	adverb, superlative	<i>fastest</i>	:	mid-sentence punc	<i>: ; ... --</i>
RP	particle	<i>up, off</i>			

Natural Language Processing

Hearst Patterns

■ Hearst-Patterns (1992) [Marti Hearst]

- Lexico-syntactic patterns
- is a, such as, including,...
- E.g. Java is an object-oriented programming language



■ Marti Hearst is also active in => Search User Interfaces



Social Media Mining

Microblog Peculiarities

■ Twitter => X

- 280 Characters
- Language peculiarities
 - Sentence grammar not given
 - Emoticons
 - Hashtags
 - etc.



Umweltnachrichten @_umweltnews_

Flut in Südspanien: Mehrere Menschen sterben bei
Überschwemmung - FAZ - Frankfurter Allgemeine Zeitung
bit.ly/TSFLnv #umwelt

Schließen ↩ Antworten ↻ Retweeten ★ Favorisieren

9:05 PM - 29 Sep 12 · Details

■ Application Domains

- Opinion mining, detection of opinion leaders
- Analysis of communities
- „Sensors“: UGC (user generated content) as sensors („citizen sensors“)
 - E.g., crisis detection, prediction of elections

Social Media Mining

Microblog Peculiarities

■ Techniques

- Hashtag extraction
- Topic extraction
- Semantic extraction

■ Challenges

- Subset of data (e.g. tweets)
- Quality factors: trustworthiness
- Temporal data



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Selected Topics and Applications

Opinion Mining & Sentiment Analysis

Introduction & Terminology

Motivation: “**What other people think**”

- for the first time in human history: a huge volume of opinionated data recorded in digital forms
- research field related to information retrieval/extraction, NLP, data mining; established ~2001



Aaron Ruper ✓ @atruper · Oct 9, 2020

"In Europe, you go there, they have forest nations, they call them **Austria**, different places. They don't have the woods burning down, and their trees are more explosive than ours, meaning explosive from potential fire." -- Trump to Rush Limbaugh on **climate change**



What's happening to our world scares the hell out of me, but if I shout the brutal, unvarnished truth from the rooftops, will this really galvanize you and others into fighting for the planet and your children's futures?

Bill McGuire

<https://edition.cnn.com/2024/03/07/opinions/climate-scientist-scary-doom-anxiety-mcguire>

Opinion Mining & Sentiment Analysis

Introduction & Terminology

⇒ Definition/difference of terms **opinion mining** and **sentiment analysis** varies

- **Opinion Mining** [coined 2003] is the computational treatment of opinion, sentiment, and subjectivity in text [Pang & Lee 2008].
- **Sentiment Analysis** [coined 2001]: also known as **opinion mining**: to understand the attitude of a speaker or a writer with respect to some topic [Wei 2010]

'The beginning of wisdom is the definition of terms,' wrote Socrates. The aphorism is highly applicable when it comes to the world of social media monitoring and analysis, where any semblance of universal agreement on terminology is altogether lacking.

Today, vendors, practitioners, and the media alike call this still-nascent arena everything from 'brand monitoring,' 'buzz monitoring' and 'online anthropology,' to 'market influence analytics,' 'conversation mining' and 'online consumer intelligence'.

Jeff Zabin and Alex Jefferies. Social media monitoring and analysis: Generating consumer insights from online conversation. Aberdeen Group Benchmark Report, January 2008; cited by (Pang & Lee 2008)

Opinion Mining & Sentiment Analysis

Definitions

Opinion Mining

- a subjective (or objective) judgement, e.g., “the size of the camera is adequate”, “the resolution of the camera is great”, “the camera broke within a week”
- Result: (camera size, great) (camera resolution, great) but also (camera size, +)

Sentiment Analysis

- a feeling, e.g., “I feel sad about my camera”, “I am happy about the election results”
- Result: (-), (+) – often sentence level

Opinion: a quintuple: entity, aspect (feature), orientation (polarity), opinion holder, time

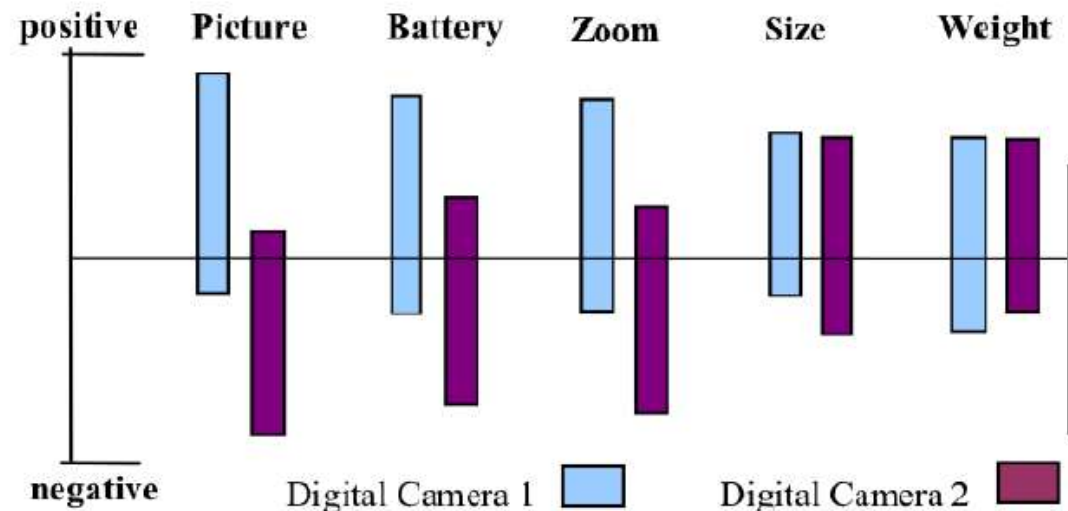
Opinion orientation: can be positive, negative or neutral, or be expressed with different strength/intensity levels [Liu 2012]

More related concepts/terms: **subjectivity**, **emotions**, **irony**, etc.

Opinion Mining & Sentiment Analysis

Sources, Domains, Applications

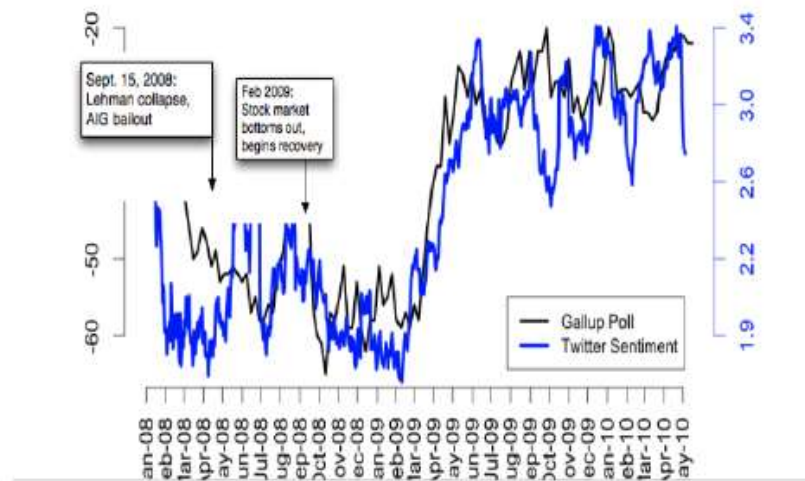
- **Sources**
 - Web blogs, Twitter messages, reviews (product, hotel, etc.)
- **Domains**
 - eBusiness, tourism, health, politics, environmental disasters, etc.
- **Applications**
 - positive/negative opinion of product features, election predictions, etc.
- **Example:** OpinionObserver



Opinion Mining & Sentiment Analysis

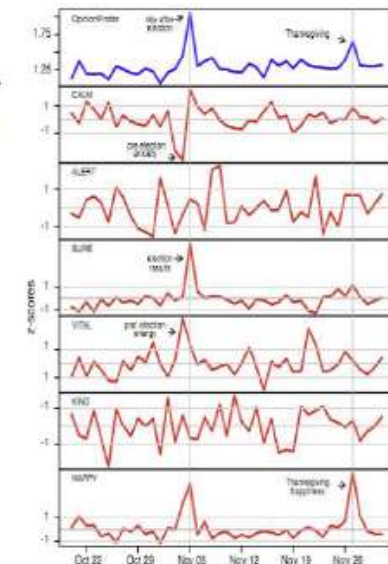
Applications

Twitter sentiment versus Gallup Poll of
Consumer Confidence
window = 15, $r = 0.804$



Twitter sentiment

Johan Bollen, Huina Mao, Xiaojun Zeng. 2011.
[Twitter mood predicts the stock market](#),
Journal of Computational Science 2:1, 1-8.
10.1016/j.jocs.2010.12.007.



[Wei 2010]

Opinion Mining & Sentiment Analysis

Sources

Source formats:

■ Structured, unstructured, mixed

257 von 266 Kunden fanden die folgende Rezension hilfreich:

★★★★☆ **Klein und fein!**

Das Wildfire S ist das Nachfolgemodell des sehr erfolgreichen und beliebten Wildfire von HTC, welches nun in einigen Punkten aktualisiert wurde.

Änderungen des "Wildfire S" gegenüber dem "Wildfire"

- die Auflösung des Bildschirms wurde wesentlich verbessert (480x320 Pixel statt 320x240)
- es wurde auf die...

[Vollständige Rezension lesen >](#)

Vor 7 Monaten von Chevin veröffentlicht

+ Was uns gefällt

- + großes Display
- + günstig
- + bringt die meisten Funktionen eines High-End-Smartphones von Samsung mit

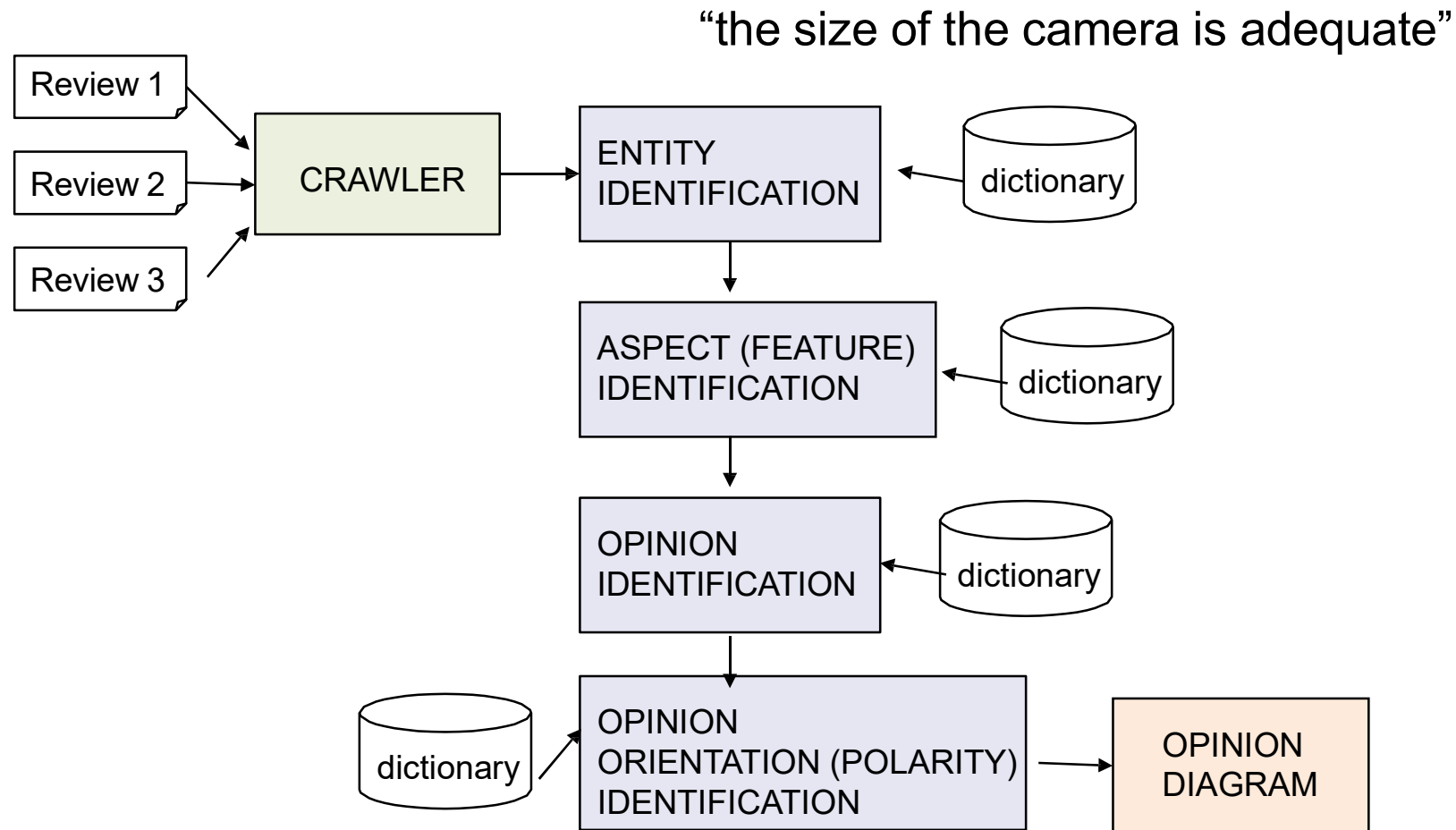
- Was uns nicht gefällt

- sowohl ein echtes Tablet als auch ein echtes Smartphone erscheinen uns sinnvoller

Wir meinen:

Samsung bringt hier die Antwort auf den iPod Touch - allerdings mit ein paar Jahren Verspätung. Im Vergleich zu günstigen Tablets oder Android-Smartphones ist das Galaxy S WiFi 5.0 weder Fisch noch Fleisch. Keine Frage, das Gerät ist gut - aber wohl nur für wenige wirklich ein sinnvoller Kauf.

Opinion Mining & Sentiment Analysis Process



Opinion Mining & Sentiment Analysis

Identification of Features

Types of features

- **Explicit features:** explicitly given in text
 - ◆ Example: good manual → feature <manual>
- **Implicit features:** have to be inferred from text
 - ◆ Example: included 16MB is stingly → feature <memory>

Feature identification techniques

- **Knowledge engineering:** based on a dictionary, ontology etc.
- **Statistic Approaches:** e.g. frequent terms
- **Machine Learning:** e.g. manual annotation of learning set

Opinion Mining & Sentiment Analysis

Identification of Opinions

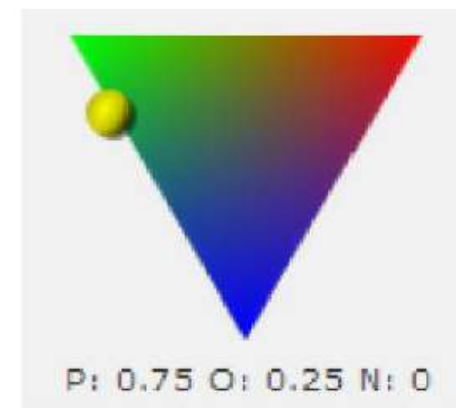
Extraction of opinion which is related to a feature

- Opinion is often expressed by an **adjective**
- Dictionaries containing opinion orientations exist, e.g. Sentiwordnet

Sentiwordnet

- Contains terms and a **positive** and/or **negative weight**
- Based on **Wordnet**
- Available in **English** and **German**

Is „cold“ negative or positive?



Opinion Mining & Sentiment Analysis

Challenges

■ Granularity

ABSA: aspect oriented sentiment analysis

◆ document => sentence => aspect level

☞ „room was clean but breakfast was ugly“

☞ „my cold and my severe headache is getting better“

Opinion Mining & Sentiment Analysis

Examples (*) and Challenges

■ Sentiments without opinion terms

- ◆ „Die Zimmer haben alle einen Teppichboden, der Geschichten erzählt,, (negative)
- ◆ „Der Teppich stammt laut den Flecken aus dem Jahr 1950“ (negative)

[* aus Masterarbeit Omid Sabri, 2012]

References

- Bo Pang, Lillian Lee: Opinion mining and sentiment analysis. Foundations and Trends in Information Retrieval 2008
<https://www.cs.uic.edu/~liub/FBS/SentimentAnalysis-and-OpinionMining.pdf>
- Bing Liu: Sentiment Analysis and Opinion Mining, 2012
<https://www.cs.uic.edu/~liub/FBS/SentimentAnalysis-and-OpinionMining.pdf>
- Furu Wei: Sentiment Analysis and Opinion Mining, Slides IIR, 2010
<http://web.nchu.edu.tw/~jodytsao/MarketingG/IIR10-Sentiment%20Analysis.pdf>
- Walaa Medhat et al: Sentiment analysis algorithms and applications: A survey, 2014
<http://www.sciencedirect.com/science/article/pii/S2090447914000550>
- Bing Liu: Opinion Observer: Analyzing and Comparing Opinions on the Web, WWW2005

Ali Reza Alaei, Susanne Becken, and Bela Stantic. 2019. Sentiment Analysis in Tourism: Capitalizing on Big Data. en. *Journal of Travel Research*, 58, 2, (February 2019), 175–191. ISSN: 0047-2875, 1552-6763. DOI: 10.1177/0047287517747753. Retrieved 12/28/2022 from <http://journals.sagepub.com/doi/10.1177/0047287517747753>.

C. Hutto and Eric Gilbert. 2014. VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8, 1, (May 2014), 216–225. ISSN: 2334-0770, 2162-3449. DOI: 10.1609/icwsm.v8i1.14550. Retrieved 01/01/2023 from <https://ojs.aaai.org/index.php/ICWSM/article/view/14550>.